

Mechanical Reasoning Field Guide

Pulleys · Gears · Levers · Hydraulics · Centre of gravity · Beams, with the rest of the classical statics and dynamics that mechanical-reasoning tests draw on

0 The one idea

every machine is a force–distance trade

Golden rule (work in = work out):

$$F_{\text{effort}} d_{\text{effort}} = W_{\text{load}} d_{\text{load}}$$

So $MA = \frac{W}{F} = \frac{d_{\text{effort}}}{d_{\text{load}}}$ (ideal, frictionless)

If you can say *how far each end moves*, you have the answer, whatever the machine. A machine that makes force *smaller* makes you move *farther*, by exactly the same factor. Nothing multiplies energy.

Four conservation laws do all the work

Forces $\sum F = 0$ in equilibrium (up = down, left = right)

Moments $\sum Fd = 0$ about *any* point you like

Energy $Fs_{\text{in}} = Fs_{\text{out}}$; $mgh \leftrightarrow \frac{1}{2}mv^2$

Volume incompressible fluid: $A_1s_1 = A_2s_2$

Recognise the configuration in one glance

You see	Law to use	One-line answer
ropes, wheels, load	count strands on the movable block	$F = W/n$
circles with teeth	ratio telescopes; sense alternates	$\omega_{\text{out}} = \omega_{\text{in}} T_{\text{in}}/T_{\text{out}}$
bar on a triangle	moments about the pivot	$\sum w_i(x_i - p) = 0$
two pistons	Pascal: equal pressure	$F_2 = F_1 A_2/A_1$
open vessel, dots	depth only	deepest dot wins
stacked blocks, ledge	centre of mass vs. support edge	COM past edge $\Rightarrow$ tips
bar on two supports	moments about the other support	$R_A = \sum W_i(L - x_i)/L$

The 20-second protocol

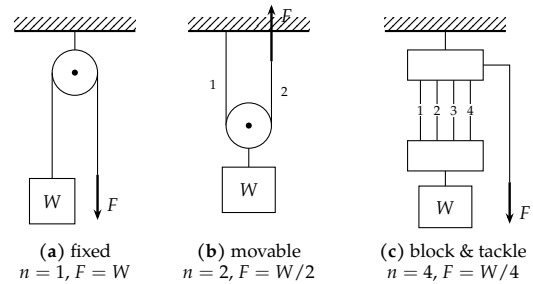
- Name the configuration** (table above). This picks the law.
- Predict the direction first:** bigger or smaller than the input? Same or opposite sense? This alone kills half the options.
- Compute once** with the one-liner.
- Check by a second route:** energy for pulleys and pistons, moments about a second point for levers and beams, stage by stage for gears.
- Match the traps:** if an option equals the input, the inverse ratio, or double or half your answer, it is a planted mistake, not a coincidence.

1 Pulleys

count the strands

$MA = n =$ number of rope segments **holding up the movable block**. $F = W/n$.

Energy check: raise the load $h \Rightarrow$ each of the n strands shortens by $h \Rightarrow$ you pull nh of rope. $F \cdot nh = W \cdot h$.



How to count. Draw an imaginary horizontal line just *above* the movable block and count the strands it cuts. The free end you pull *down* on from a fixed wheel is *not* counted (it only changes direction, as in (a)). If the free end leaves the *movable* block pulling *upward* (as in (b)), it is counted.

- Rope end tied to the *movable* block $\Rightarrow n$ is odd; tied to the *fixed* block $\Rightarrow n$ even.
- Rope speed = $n \times$ load speed. Weight of the movable block adds to W if the problem says it has one.

Example. $n = 4, W = 240 \text{ N} \Rightarrow F = 60 \text{ N}$; lifting 1 m needs 4 m of rope ($60 \times 4 = 240 \times 1$).

Traps built into the options: $F = W$ (ignores MA); $W/(n \pm 1)$ (mis-counted strands); $W/2$ ("every tackle halves"); $W/2n$ (counts each wheel twice); nW (multiplied instead of divided).

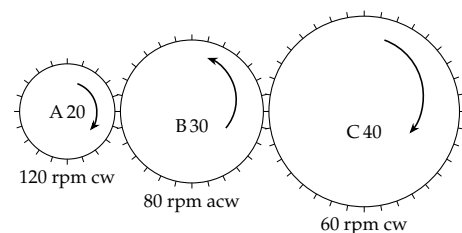
2 Gears and belts

ratios telescope, sense alternates

Speed: teeth $\times$ rpm is the same for meshing gears: $T_1\omega_1 = T_2\omega_2$.

Simple train: idlers cancel. $\omega_{\text{last}} = \omega_{\text{first}} \cdot T_{\text{first}}/T_{\text{last}}$.

Sense: reverses at every external mesh. k meshes: k even $\Rightarrow$ same sense as driver.

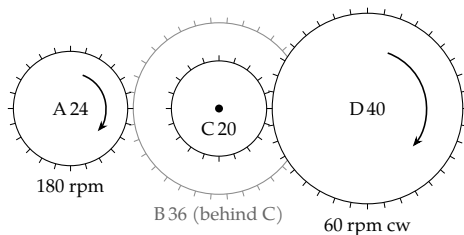


Example. A (20 T, 120 rpm cw) $\rightarrow$ B (30 T) $\rightarrow$ C (40 T). $\omega_C = 120 \times 20/40 = 60 \text{ rpm}$. B's 30 never enters. Two meshes $\Rightarrow$ C turns **clockwise**. Fast sense check: *adjacent gears always turn opposite ways*; odd-numbered gears (1st, 3rd, 5th) share the driver's sense.

Compound gears (two wheels on one shaft)

Wheels fixed to the same shaft share the same *rpm*, not the same teeth rate. Multiply stage ratios:

$$\omega_D = \omega_A \cdot \frac{T_A}{T_B} \cdot \frac{T_C}{T_D} = \omega_A \frac{\text{product of drivers}}{\text{product of driven}}.$$



Example. $\omega_D = 180 \times \frac{24}{36} \times \frac{20}{40} = 180 \times \frac{480}{1440} = 60 \text{ rpm}$. Meshes: A-B and C-D (the shared shaft is not a mesh) $\Rightarrow$ 2 meshes $\Rightarrow$ D turns **clockwise** like A.

Belts, chains and other drives

- **Belt/chain:** same rim speed $\Rightarrow r_1\omega_1 = r_2\omega_2$ (radius or sprocket teeth replace gear teeth). **Open** belt keeps the sense; **crossed** belt reverses it.
- **Internal (ring) gear:** pinion and ring turn the *same* way.
- **Rack and pinion:** rack speed = $r\omega$; one turn moves the rack T teeth.
- **Worm:** one turn of a single-start worm advances the wheel one tooth $\Rightarrow$ ratio = T_{wheel} .
- **Torque** goes the other way: $\tau_{\text{out}} = \tau_{\text{in}} T_{\text{out}}/T_{\text{in}}$ (slower = stronger).
- **Meshing sanity:** three gears in a closed triangle cannot turn at all (each pair must reverse).

Traps built into the options: right speed, wrong sense; ratio inverted ($\omega T_{\text{last}}/T_{\text{first}}$); first mesh only; “idlers cancel the speed too” (answer = input rpm); for compound trains, treating B/C as one idler ($\omega T_A/T_D$), stopping after stage one, or pairing teeth wrongly ($T_A T_B/T_C T_D$).

3 Levers and moments

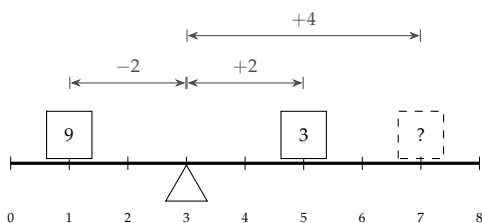
moments about the pivot

Moment = force $\times$ perpendicular distance to the pivot.

Balance: $\sum_i w_i (x_i - p) = 0$

(signed distances: left of pivot negative, right positive). Un-

known at x_u : $W = -\frac{\sum w_i (x_i - p)}{x_u - p}$.



Example. Pivot at 3. $9(1-3) + 3(5-3) + W(7-3) = 0 \Rightarrow -18 + 6 + 4W = 0 \Rightarrow W = 3 \text{ N}$. Check about mark 7: pivot reaction $R = 9 + 3 + 3 = 15 \text{ N}$; $9(-6) + 3(-2) + 15 \cdot 4 = -54 - 6 + 60 = 0 \checkmark$.

Speed tip. Keep a two-column tally: anticlockwise moments | clockwise moments. Only the *difference* matters; divide it by the unknown's arm.

Classes of lever (what sits in the middle)

- | | | |
|------------|----------------|--|
| 1st | fulcrum middle | seesaw, crowbar, scissors: $MA \geq 1$ |
| 2nd | load middle | wheelbarrow, nutcracker, bottle opener: $MA > 1$ |
| 3rd | effort middle | tweezers, forearm, fishing rod: $MA < 1$, gains speed |

$MA = \text{effort arm} / \text{load arm}$. A longer handle (spanner, door handle far from the hinge) needs less force for the same moment.

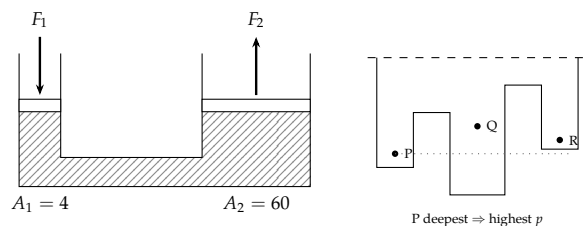
Traps built into the options: equating *weights* instead of moments; measuring the arm from the wrong tick (± 1 mark); taking moments about the beam's *end* instead of the pivot; $2W$; $W + 1$.

4 Hydraulics and fluid pressure pressure is shared, depth decides

Pascal: pressure is the same throughout a connected fluid: $\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_2 = F_1 \frac{A_2}{A_1}$.

Volume: $A_1 s_1 = A_2 s_2 \Rightarrow s_2 = s_1 \frac{A_1}{A_2}$ (big piston moves less).

Depth: $p = p_0 + \rho gh$; depends on depth h only, never on width or shape.



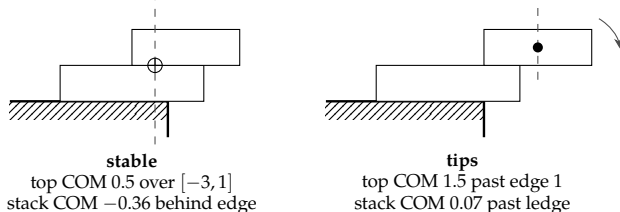
Example. $A_1 = 4 \text{ cm}^2$, $A_2 = 60 \text{ cm}^2$, $F_1 = 50 \text{ N}$; $p = 12.5 \text{ N/cm}^2$, $F_2 = 750 \text{ N}$. Push the small piston in 15 cm: $s_2 = 15 \times 4/60 = 1 \text{ cm}$. Energy: $50 \times 15 = 750 \times 1 \checkmark$.

- **Diameters, not areas?** $A \propto d^2$: doubling the diameter quadruples the force ratio.
- **Hydrostatic paradox:** a narrow tall column and a wide one of equal height press equally on their bases; a “narrow neck” or “wide chamber” above a point is irrelevant. Only the vertical drop from the free surface counts. Points at the same depth in one connected fluid are at the same pressure.
- **Water:** $\approx 10 \text{ kPa}$ per metre ($\approx 1 \text{ atm}$ per 10 m).
- **Buoyancy** (Archimedes): upthrust = weight of fluid displaced. Floats when $\rho_{\text{obj}} < \rho_{\text{fluid}}$; floating fraction submerged = $\rho_{\text{obj}}/\rho_{\text{fluid}}$.
- **Flow:** $A_1 v_1 = A_2 v_2$; narrower pipe $\Rightarrow$ faster flow and (Bernoulli) *lower* pressure.

Traps built into the options: ratio inverted ($F_1 A_1 / A_2$, or big piston moves *farther*); force transmitted unchanged ($F_2 = F_1$: it is *pressure* that is equal); area difference $A_2 - A_1$; $2\times$. In vessels: points in wide sections or under “more water” that are shallower.

5 Centre of gravity and tipping centre of mass over the base

A body (or stack) is stable while the vertical line through its **centre of mass** (COM) falls **within its base of support**. Past the edge $\Rightarrow$ it tips about that edge. Uniform blocks of equal thickness: weight $\propto$ width, acting at the midpoint. $x_{\text{COM}} = \frac{\sum w_i m_i}{\sum w_i}$ (m_i = midpoint).



Stack test, top down:

1. COM of the top block alone must lie over the block below it.
2. COM of the top two must lie over the third, and so on.
3. COM of the whole stack must lie over the ledge (behind the edge).

Any failure $\Rightarrow$ that stack tips. *Overhang by itself proves nothing*; a block can stick far out if enough weight sits behind the edge.

Example. Lower block $[-3, 1]$ (weight 4), upper $[-1, 2]$ (weight 3), edge at 0. Upper COM = $0.5 \in [-3, 1] \checkmark$. Whole: $(4 \cdot (-1) + 3 \cdot 0.5)/7 = -0.36 < 0 \checkmark \Rightarrow$ stable.

- The biggest overhang is the lure; compute signed moments about the edge: $\sum w_i(m_i - e) > 0 \Rightarrow$ tips.
- Harmonic stacking: identical bricks of length 1 can overhang $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots$ per level.
- General stability: lower COM and wider base $\Rightarrow$ harder to tip. Tipping angle θ : $\tan \theta = (\text{half-base})/(\text{COM height})$.
- A suspended object hangs with its COM directly below the pivot.

Traps built into the options: the stack with the largest overhang; the tallest stack; the one whose *bottom* block overhangs most but carries a counterweight.

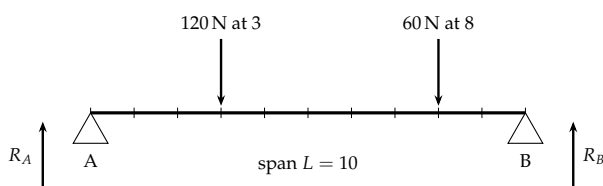
6 Beams on two supports the nearer support carries more

Take moments about one support to find the *other* reaction:

$$R_A = \frac{\sum W_i (L - x_i)}{L}, \quad R_B = \frac{\sum W_i x_i}{L}$$

check: $R_A + R_B = \sum W_i$

Each load splits in *inverse* proportion to its distances: the nearer support carries more.



Example. $R_A = (120 \times 7 + 60 \times 2)/10 = 96 \text{ N}$; $R_B = (120 \times 3 + 60 \times 8)/10 = 84 \text{ N}$; $96 + 84 = 180 \checkmark$.

Per-load shortcut. A load at fraction $f = x/L$ from A sends $(1 - f)$ of itself to A and f to B. Load at the centre: half each. Two people carrying a pole: the one nearer the load works harder.

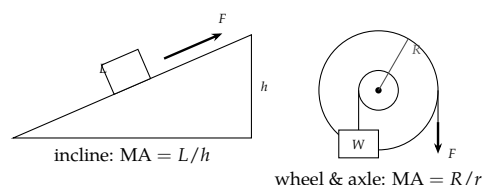
- **Uniform beam weight** acts at its midpoint (half to each support).
- **Overhanging (cantilever) load** beyond B: R_A can go *negative* (A must hold the beam *down*).

Traps built into the options: reactions swapped (arm measured from the wrong support: gives R_B); load split equally regardless of position; whole load to A; +10 slip.

7 Beyond the six

the classic test topics

Simple machines



Screw/jack $MA = 2\pi\ell/p$ (handle length ℓ , pitch p)

Wedge $MA = \text{length/thickness}$; thinner = easier

Spanner $\tau = F\ell$: double the handle, halve the force

Springs

Hooke: $F = kx$. **Series:** $1/k = 1/k_1 + 1/k_2$ (softer; each spring feels the full load, extensions add). **Parallel:** $k = k_1 + k_2$ (stiffer; load shared). Cutting a spring in half doubles k .

Motion and energy

- **Frictionless slope or slide:** speed at the bottom $v = \sqrt{2gh}$ depends only on the *drop*, not the path or the mass. A steeper path gets there *sooner*, not faster.
- **Falling:** all masses fall together (no air resistance). A ball dropped and one thrown horizontally land at the same time.
- **Projectile:** horizontal velocity constant; range max at 45° ; 30° and 60° give equal ranges.
- **Pendulum:** $T = 2\pi\sqrt{\ell/g}$; independent of mass and (small) amplitude. Longer $\Rightarrow$ slower.
- **Circular motion:** a released object flies off along the *tangent*, never outward along the radius. Same rpm $\Rightarrow$ outer point moves faster ($v = \omega r$).
- **Momentum:** in collisions $\sum mv$ is conserved; a gun recoils; a cart and person pushing apart gain equal and opposite momenta.
- **Friction:** $f \leq \mu N$; independent of contact area; kinetic $<$ static (hardest to *start* sliding).

Heat, sound, electricity

- Metals expand when heated: a bimetal strip bends *toward* the less-expanding metal; a heated ring's hole gets *bigger*.
- Dark matte surfaces absorb and radiate best; shiny surfaces reflect.
- Shorter, tighter or thinner string $\Rightarrow$ higher pitch. Sound needs a medium; it travels faster in solids.
- Series bulbs share voltage (dimmer; one fails $\Rightarrow$ all off); parallel bulbs each get full voltage.

In a well-made test every distractor is a *named mis-computation*. When your answer appears, look for its siblings; they confirm you read the right configuration.

Pattern	The mistake it encodes
= input	ignored the machine (pulley $F = W$, piston $F_2 = F_1$, gear $\omega = \omega_{\text{in}}$)
inverse	ratio upside down (gear teeth, piston areas, beam arms)
$\times 2, \div 2$	doubled or halved: the classic “every tackle halves” slip
± 1 step	strand miscount; arm from the wrong tick
right number, wrong sense	forgot gears alternate, or counted the shared shaft as a mesh
equal split	ignored position (beam load halved, lever weights equated)
biggest look	overhang or “more water above” instead of COM or depth

Final check, every problem: Does the answer move in the predicted direction? Does work in = work out? Do all the forces (and moments) balance? If all three pass, commit and move on.

- D1. Block and tackle, rope end tied to the movable block, 5 strands support it. $W = 450\text{ N}$. Effort?
- D2. Simple train A(15 T, 240 rpm cw) $\rightarrow$ B(45) $\rightarrow$ C(36) $\rightarrow$ D(20). Speed and sense of D?
- D3. Compound: A(18 T, 300 rpm cw) meshes B(60); C(40) on B’s shaft meshes D(24). D?
- D4. Beam 0–8, pivot at 4; 12 N at 1, 8 N at 6. What weight at 8 balances?
- D5. Pistons of *diameter* 2 cm and 10 cm; 40 N on the small one. Force on the large one?
- D6. Span 12 marks, one 90 N load at mark 4. Reaction at A (mark 0)?
- D7. Ledge edge at 0. Lower block $[-2, 2]$, upper block $[-1, 3]$, same thickness. Stable?
- D8. A 600 N crate is pushed up a frictionless ramp 6 m long rising 1.5 m. Force along the ramp?

Answers. D1 90 N. D2 180 rpm anticlockwise (3 meshes; B and C cancel). D3 $300 \cdot 0.0018 \cdot \frac{60}{24} = 150$ rpm clockwise. D4 $\frac{12(-3) + 8(2) + 4W}{90} = 0 \Rightarrow W = 5\text{ N}$. D5 area ratio 25 $\Rightarrow$ 1000 N. D6 $12(8/12) + 8(1/12) = 60\text{ N}$. D7 top COM 1 $\in [-2, 2]$ is fine, but whole-stack COM $(4 \cdot 0 + 4 \cdot 1)/8 = 0.5 > 0 \Rightarrow$ tips. D8 $600 \cdot 1.5/6 = 150\text{ N}$.